

Drone-Based Reservoir Surveying and Water-Intelligence Service

Project Proposal

This project proposes a drone-based reservoir surveying and water-intelligence service for government agencies.





1. Executive Summary

This project proposes a drone-based reservoir surveying and water-intelligence service for government agencies. A drone flies over a reservoir using predefined GPS/GNSS waypoints, collects water-depth data at selected points using a sonar sensing unit, and combines those measurements with reservoir boundary mapping to estimate the actual stored water volume.

The service is designed not as a hardware sale, but as a professional, contract-based government service that delivers storage estimates, depth maps, and decision-support reports for irrigation, water supply, hydropower, and disaster-management authorities.

2. Problem and Urgency

CORE PROBLEM

Authorities often know the surface water level, but not the true remaining storage volume.

- A reservoir may appear full, but usable storage may be much lower.
- Sedimentation and bed-shape changes reduce accuracy.
- Outdated storage curves lead to old volume estimates.

Existing Solutions and Their Gaps



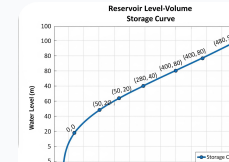
Manual Gauges

- Level monitoring
- Weak for storage



Fixed Sensors

- Continuous
- No depth data



Historical Curves

- Easy to use
- Inaccurate



Satellite

- Broad range
- Low detail

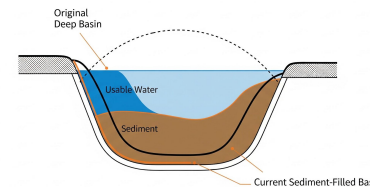


Boat Surveys

- Very accurate
- Costly/Slow

WHY SURFACE OBSERVATION IS NOT ENOUGH

- Shows only the top level, not depth variation.
- Fails to reveal underwater basin conditions.
- Does not guarantee accurate storage estimation.



WHY OLD CALCULATIONS FAIL

- Old surveys do not reflect current sedimentation.
- Sediment reduces actual capacity over time.
- Same level != Same volume due to basin changes.

OPERATIONAL IMPACTS OF POOR KNOWLEDGE

- Poor irrigation planning • Drinking-water uncertainty • Weak drought response • Flood-release mismanagement • Hydropower scheduling problems • Sector conflicts • Environmental flow risks • Poor long-term policy decisions

3. Why This Matters



Economic Impact

- Poor estimates increase operational inefficiency
- Wrong release decisions waste water resources
- Planning errors raise management costs



Agricultural Impact

- Reliable releases essential for irrigation planning
- Estimates disrupt planting schedules
- Water uncertainty increases agricultural risk



Public Water Supply

- Utilities need data for supply planning
- Poor estimates increase shortage risk
- Continuity depends on accurate reservoir knowledge



Climate Resilience

- Climate uncertainty increases planning pressure
- Data supports drought preparedness
- Intelligence improves climate adaptation



Disaster Preparedness

- Data supports emergency release decisions
- Condition affects flood-buffer planning
- Improves disaster response readiness



National Planning

- Supports national policy decisions
- Improves infrastructure planning
- Supports long-term national resilience

4. Regions Facing This Problem

Sri Lanka

Heavy dependence on reservoirs for irrigation and hydropower. Climate stress increases need for accuracy.

India

Large network with active monitoring requirements for agriculture and drought planning.

Pakistan

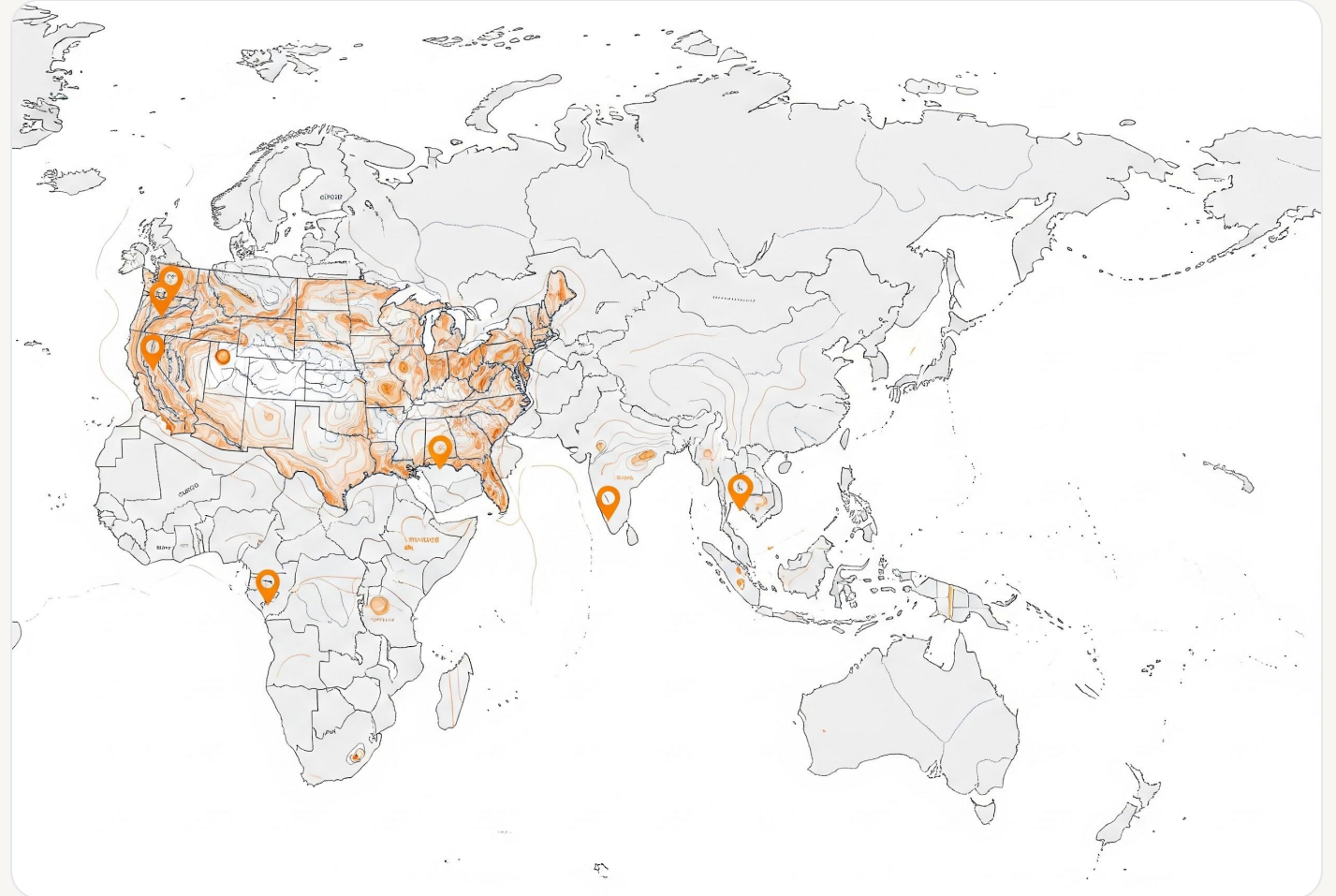
Severe shortages and inter-provincial water pressure make monitoring critical.

South Asia

High climate vulnerability and irrigation dependence across the entire region.

Mekong & Western USA

Global operational challenge from shared basins to advanced developed systems.



5. Our Proposed Service Model

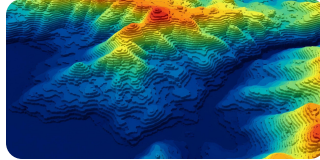
Reservoir Surveying and Water Intelligence as a Service

SURVEY MISSIONS



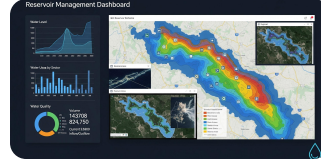
Reservoir survey missions

DEPTH MAPPING



Water-depth mapping

GIS OUTPUTS

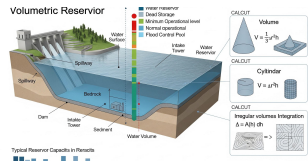


GIS outputs and dashboard reporting

Why a Service Model Is Stronger Than Hardware Sales

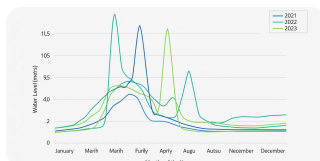
- Lower upfront cost
- No need for an in-house drone team
- No maintenance burden
- No calibration burden
- No software integration burden
- Easier procurement process
- Contract-based deliverables
- Lower operational risk for clients

STORAGE ESTIMATION



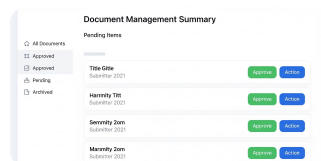
Storage-volume estimation

TREND COMPARISON



Change-over-time trend comparison

DECISION SUPPORT



Decision-support summaries for government agencies

Public-Sector Positioning

- Not a drone purchase or gadget sale
- A specialized technical information service
- Decision-support service for govt operations

Primary Government Clients

Irrigation, Water Supply, Hydropower
Disaster Management, Agriculture
Environmental & Basin Agencies
Local and Provincial Authorities

6. Commercial Viability

Why This Becomes a Business

- Reservoirs require repeated monitoring
- Government need is continuous
- Multi-agency service value
- Scalable to national levels

Revenue Models

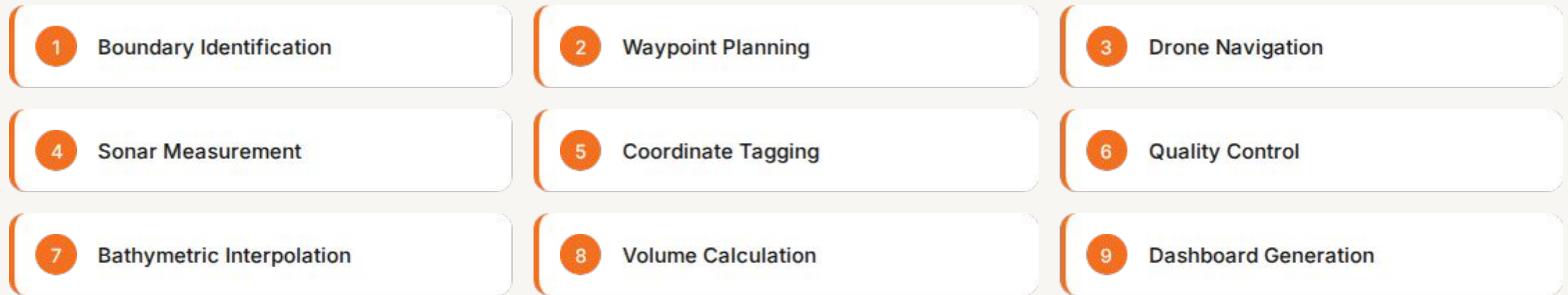
- Per-survey & Annual contracts
- Emergency drought/flood packages
- Dashboard subscription fees
- Regional framework contracts

Business Value

- Optimized water allocation decisions
- Reduces dependency on old data
- Recurring service revenue streams
- Long-term government contracts

7. Technical Architecture and Workflow

Mission Workflow



Likely Technology Stack

ArduPilot

ROS 2

SLAM Toolbox

RTK / GNSS

GIS Mapping

Core Outputs

- Current storage estimate & Depth map
- Usable-water estimate & Trend analysis
- Decision-ready condition reports

8. Feasibility & Practical Challenges

Technical Feasibility

Technically possible with current drone, sonar, and GIS tools. Strongest use case: periodic surveys for decision support.



Environmental

Wind, weather, battery limits



Technical

Signal loss, positioning accuracy



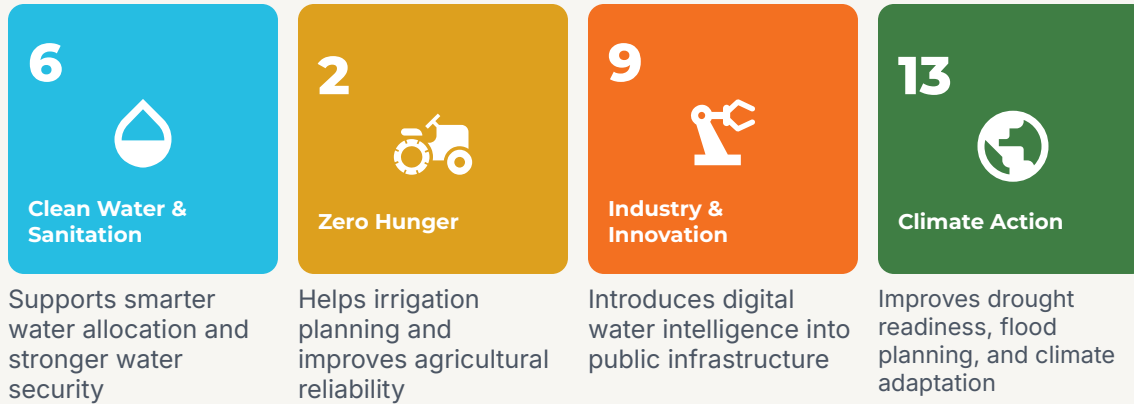
Regulatory

Permissions and aviation safety

Mitigation Strategies

- ✓ Use structured flight planning
- ✓ Use RTK for georeferencing
- ✓ Apply QA/QC during processing
- ✓ Validate with manual measurements
- ✓ Start with controlled pilot sites
- ✓ Include approval timelines in planning

9. Sustainability & SDG Alignment



10. Risks and Limitations

- Technical:** Measurement error / interpolation uncertainty.
- Institutional:** Slow procurement / agency coordination.
- Regulatory:** Aviation compliance / safety procedures.

11. Pilot Plan

Pilot Scope: 3–5 Reservoirs

One major, one medium, and one multipurpose reservoir under one authority in Sri Lanka.

Success Metrics

- Depth consistency
- Repeatability
- Accuracy improvement
- Turnaround time

Scale-Up Pathway



12. Final Pitch

This is not just a student project and not just a drone device.
It is a smart reservoir intelligence service for governments.

The Real Value

Accurate storage estimates & updated depth intelligence

Repeatable monitoring & decision-ready reports

Target Agencies:

- Irrigation & Drinking Water
- Hydropower Generation
- Disaster Management

By helping authorities understand water availability and reservoir conditions, we support **better planning, better allocation, and stronger climate resilience.**

Thank You